A Studies and Analyses

STUDIES 1A/1B

Slider ratings

Participants compare two complete descriptions

STUDY 2

Production

Participants type their own descriptions

Behavioural models

Order preferences across referential contexts

RSA production models

Full distribution of the fifteen possible descriptions

Controlled simulation

When does an order help the listener?

Held-out prediction

How well each model predicts each response

B Size Semantics × Degree of Planning

SIZE SEMANTICS

Context-fixed

big ← whole scene

big is judged against the whole scene.

Sequential context updating

big ← blue objects ← blue + N

big is judged among the blue objects.

DEGREE OF PLANNING

more planning ⟵ ⟶ less planning

Global
(fully planned)

κ = 0

big blue N

Complete descriptions are evaluated as wholes.

Plan-guided

0 < κ < 1

value of the complete description

big → blue → N

The complete description guides each adjective.

Fully incremental

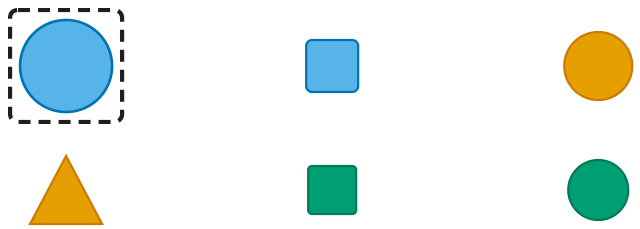
κ = 1

big → blue → N

Each adjective is compared with those available at that point.

All models assign probabilities to the same fifteen descriptions.

C Referential Task



Six objects · target in the dashed frame
Here, size alone identifies the target.

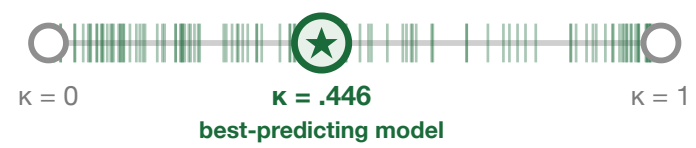
POSSIBLE DESCRIPTIONS

big | *blue* | *big blue* | *blue big* | *big blue round* | …

One to three adjectives in any order:
fifteen ordered categories of size, colour, and form

D Main Findings

DEGREE OF PLANNING



Each tick is one speaker's estimated κ: speakers differ in how much they plan.

SIZE SEMANTICS

Fitted production

Context-fixed predicts better than updating

Controlled simulation

Size-first advantage under both semantics

An intermediate degree of planning, varying across speakers, best predicts which descriptions speakers produce.